

AITEE HUB: CYBERSECURITY CAPSTONE 1 HEALTHCARE SECURITY ASSESSMENT

AITEE Community Health Center is a rapidly growing healthcare provider operating with 50 employees (doctors, nurses, administrative staff) and a small internal IT team. Their technical setup includes patient registration software, Electronic Health Records (EHR), staff laptops, Wi-Fi networks, corporate email, and a basic server environment.

Domain.

Observe vulnerability/current state

1.password	Employees use simple, easy-to-guess passwords and reuse credentials across personal and work accounts
2.Authentication (MFA)	Multi-Factor Authentication (MFA) is not enforced on critical systems or administrative accounts
3.User Awareness	Employees have received no official cybersecurity awareness or phishing simulation training.

Task 1: Asset Identification & CIA Triad

No. Asset name. Why Is It Important. What Could Happen If
Lost/Compromised

1.Patient Records	Contains sensitive medical history and personal information	Privacy breach, legal penalties, loss of patient trust.
2.Electronic Health Record (EHR) System	Stores and provides access to patients' medical information	Doctors may be unable to access records, causing delays in treatment.
3.Staff Email Accounts	Used for communication and sharing important health-center information	Account takeover, data theft, phishing attacks and unauthorized access.

4. Medical Devices	Used to monitor and support patient diagnosis and treatment. Service disruption, incorrect readings and possible harm to patients.	Service disruption, incorrect readings and possible harm to patients.
5. Backup data	Provides copies of important patient and business information for recovery.	Permanent data loss and difficulty recovering from ransomware or system failure.

TASK 2: THREAT IDENTIFICATION

No.	Threat vector	What System / Asset Could It Affect?	Possible Impact / Consequence
1.	Ransomware	EHR system, computers and patient records	Data becomes inaccessible, disrupting healthcare services.
2.	Phishing Emails	Employee email accounts and laptops	Credentials stolen and unauthorized network access gained.
3.	Insider Threat	Patient records and EHR system	Sensitive information may be accessed, changed or stolen
4.	Device Theft	Staff laptops and computers	Confidential patient data could be exposed.
5.	Power Outage	Servers, computers and medical devices	Systems become unavailable and healthcare operations are disrupted.

TASK 3: VULNERABILITIES

These are taken directly from the 8 security audit findings in your scenario.

No.	Vulnerability (Weakness)	Related Threat Vector	Why Is It Dangerous?
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1.Simple and Reused Passwords	Account Compromise / Phishing	Attackers can easily guess or use stolen credentials to access accounts.
2.No Multi-Factor Authentication (MFA)	Phishing / Account Compromise	A stolen password may be enough for an attacker to access critical systems
3.No Cybersecurity Awareness Training	Phishing	Employees may not recognize phishing emails and other cyber threats.
4.Irregular Patching and Updates	Malware / Ransomware	Unpatched systems may contain known vulnerabilities that attackers can exploit
5.Unencrypted Medical Data	Data Breach / Unauthorized Access	Sensitive patient information can be easily exposed if accessed by unauthorized people

TASK 4: DEFINE 5 SECURITY RISKS

No. Identified Risk Scenario	What Systems/Data Are Affected?	Why Does It Matter? (Business Impact)
1.Phishing attack compromises an employee account	Corporate email, staff laptops and EHR	Could lead to unauthorized access and theft of sensitive information.
2.Ransomware exploits unpatched systems	EHR, laptops and shared network drives	Could make patient records unavailable and stop normal healthcare operations
3.Unauthorized employee accesses sensitive files	Patient records and shared network drives	Could cause privacy breaches, legal problems and loss of patient trust.
4.Medical data is exposed because it is unencrypted	Electronic medical records and network drives	Attackers who gain access could read sensitive patient information.
5.Backup cannot restore data after an attack	Backup data and patient records	Data may be permanently lost, causing major

Backup data and patient records		operational disruption
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TASK 5: PRIORITIZE TOP 3 CRITICAL RISKS

No.	Top Risk Scenario	Likelihood.	Impact	Priority Level
1.	Phishing compromises employee credentials	High.	High	Critical
2.	Ransomware attacks unpatched systems	High.	High	Critical
3.	Unauthorized access to unencrypted patient data	Medium.	High	High

Justification

These three risks should be addressed first because they can have a direct and serious impact on patient information and healthcare operations. Phishing can give attackers access to employee accounts, while ransomware can make important medical records unavailable. Unencrypted patient data can also result in a serious privacy breach. Addressing these risks will greatly improve the confidentiality, integrity and availability of AITEE Health Center's information.

TASK 6: RECOMMEND SECURITY CONTROLS

No.	IDENTIFIED RISK	CONTROL TYPE	RECOMMENDED CONTROL	How IT REDUCES RISK
1.	Ransomware	Technical / Admin	Regular backups, endpoint protection	Allows recovery of data and reduces

		and security updates	malware infections.
2.Phishing / Stolen Credentials	Technical / Admin	Enforce MFA, strong password policy and phishing awareness training	Makes account takeover harder and helps employees identify phishing attempts.
3.Stolen Laptop / Data Exposure	Technical / Physical	Encrypt devices and use strong device access controls	Prevents unauthorized people from reading stored patient information.
4.Unpatched Systems	Technical / Admin	Establish regular patching and vulnerability-management procedures	Removes known security weaknesses that attackers could exploit.
5.Backup Failure	Technical / Admin	Maintain regular offline/cloud backups and test restoration	Ensures important data can be recovered after ransomware or system failure

TASK 7: EXECUTIVE SUMMARY

Executive Summary

The security assessment of AITEE Health Center identified several critical assets, threats and vulnerabilities that could affect patient safety, data confidentiality and business operations. Important assets include patient records, the Electronic Health Record system, staff email accounts, medical devices and backup data.

The major threats identified include ransomware, phishing, insider threats, device theft and power outages. Key vulnerabilities include weak passwords, lack of Multi-Factor Authentication, unpatched systems, unencrypted data and inadequate backup testing. The three highest-priority risks are ransomware attacks, phishing-related credential theft and stolen devices exposing patient information.

To reduce these risks, AITEE Health Center should implement MFA, strong password policies, regular employee security awareness training, data encryption, timely software updates and

reliable tested backups. These controls will improve the confidentiality, integrity and availability of the health center's systems and data.